

MPG-6099



Multi-parameters Water Quality Analyzer

Model : MPG-6099



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Chapter I Product Description

1.Overview

The multi-parameter water quality online analysis system integration platform can directly integrate various water quality online analysis parameters into a complete machine, and centrally view and manage it on the touch screen panel display; the system integrates water quality online analysis, data remote transmission, database and analysis. The integration of software and system calibration functions provides great convenience for modern water quality data collection and analysis.

Flexible configuration and combination, the machine can be flexibly selected (note: the specific parameters are subject to the actual order parameters): such as pH, temperature and other conventional parameters should be combined or expanded according to the actual needs of different customers; it can also be combined with on-site PLC, Central control room for data connection.

2.Features

2.1 Personalized customized combination of parameters, which can flexibly combine, select and customize corresponding monitoring parameters according to customer's needs;

2.2 Realize intelligent online monitoring application through flexible configuration of intelligent instrument platform software and combined parameter analysis module;

2.3 It has automatic online sensors, reduces manual maintenance cycles, creates a good operating environment for parameter measurement, integrates and simplifies complex on-site problems, and eliminates uncertainties in the application process;

2.4 Optional remote data link. As the old Chinese saying goes: let customers strategize and win thousands of miles away. (optional)

3.Parameters

Model		MPG-6099
Measuring parameters		pH/Temp/COD/NH4/TSS
Measuring range	Temp	0-60°C
	pH	0~14pH
	NH4	0-1000mg/L

Multi-parameter Water Analyzer

	TSS	0-45000mg/L
	COD	0-500mg/L
Resolution & accuracy	Temp	Resolution:0.1℃ Accuracy:±0.5℃
	pH	Resolution:0.1pH Accuracy:±0.1pH
	NH4	Resolution:0.01mg/L Accuracy:±5%FS
	TSS	Resolution:0.01mg/L Accuracy:±3%FS
	COD	Resolution:0.01mg/L Accuracy:±5%FS
Communication	RS485	
Power supply	220V	
Working temp	Temp: (0-50) ℃;	
humidity	Relative humidity: ≤85% RH (no condensation)	

Chapter II Product Appearance and Structure

Appearance:

Touch screen: real-time display of monitoring parameters, and can be operated by touch.

The front panel is equipped with a USB interface for Condwnloading data.

The instrument is equipped with: pH, dissolved oxygen, Turbidity electrodes, etc. (Note: The number of electrodes is subject to the customer's actual order, and the shape is subject to the actual goods received)

Note: There are corresponding labels on all electrode wires, and the labels correspond to the labels below the terminals. When wiring, just connect the corresponding terminals. Pay attention to the positive and negative poles when wiring the electrode power supply.

Chapter III Software Interface Operation Instructions

Overview:

The software interface operating system includes: main page, real-time curve, historical data, calibration and other operation items.

After the device is powered on, it automatically enters the main measurement interface - the real-time data interface.

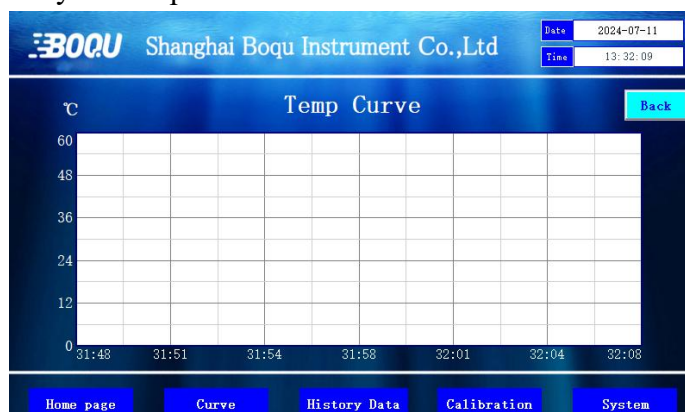
The real-time data interface displays real-time measurement data, which can clearly and intuitively reflect various indicators of current water quality, allowing customers to promptly and accurately understand the current status of various water quality parameters and conduct scientific management. as the picture shows:

Multi-parameter Water Analyzer



Click "Real-time Curve" to display the real-time curve. Click the corresponding parameter button on the left to view other parameter curves.

The real-time curve is a change curve table of the current data measured by the equipment, which can clearly and intuitively reflect the changes in various current water quality parameters, thereby facilitating customers to make adjustments to the front-end water treatment equipment according to the changes in water quality. as the picture shows:



Click "Historical Data" to view data records:

Historical data is a record of the measurement data for a period of time before the current measurement data of the equipment. This data provides a reliable guarantee for users to grasp the data measured by the equipment under unmanned operation and supervision, and also facilitates customers to query the water quality changes in a specified time period. Strong guarantee.

The screenshot shows the 'History data' interface. It displays a table with columns: Number, Times, Temp, pH, COD, TSS, and NH4. The first row shows data for 2024-07-11 13:32:07. Below the table, there are two buttons: 'History Curve' and 'Backup'. At the bottom, there are five buttons: Home page, Curve, History Data, Calibration, and System.

Number	Times	Temp	pH	COD	TSS	NH4
1	2024-07-11 13:32:07	0.00	0.00	0.00	0.00	0.00

In the "Historical Data" interface, click "Data Backup", the data export interface will pop up, you can insert a USB flash drive, select the starting time to export the data, and click the corresponding export button to export the data to the USB flash drive.

Button function description:

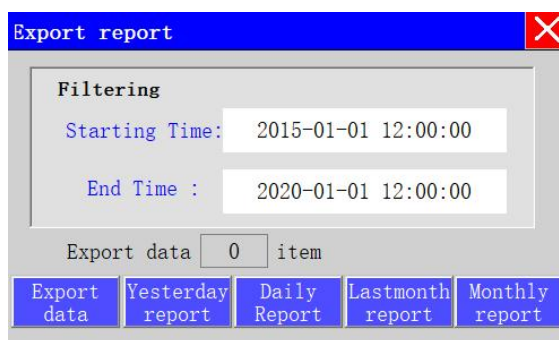
Export Data: Export all data to a USB flash drive.

Yesterday's report: Export yesterday's data to a USB flash drive.

Report of the day: Export the data of the day to a USB flash drive.

Last month report: Export the data of the last month to a USB flash drive.

Current Month Report: Export this month's data to a USB flash drive.



Click "Historical Curve" on the main toolbar to view historical data curves:



The yellow list area on the left: displays the specific parameter data and recording time of the current cursor position, as shown in the figure below:



Function description of historical curve button:



Scroll the curve one page backward (left end of X axis),



Scroll the curve half page backward (left end of X axis),



Scroll backward (left end of X axis) one main line position,



Scroll forward (right end of X-axis) one main dash position,



Scroll the curve half page forward (right end of X axis),



Scroll the curve forward (the right end of the X axis) by one page,



Set the curve start point time.



button to select a time period and view the historical data curve.

Click the system menu on the main toolbar to enter the time setting, you can change the date and time:

The first line sets the year, month, and day, and the second line sets the hours, minutes, and seconds. When the input is complete, click "Confirm" to save.

Chapter IV Calibration and setup

Instrument calibration:

Click the "Calibration" button on the main toolbar of the touch screen, enter the password "7320", and the calibration selection interface will appear:



1.pH calibration:



pH calibration steps:

Enter the pH calibration interface and set "Standard Liquid Type" to 1:

Standard liquid type description:

0: represents 4.00, 7.00, 10.00 standard solution (international standard).

1: Represents 4.00, 6.86, 9.18 standard liquid (Chinese standard).

Zero point calibration: Put the electrode into the 7.00 or 6.86 standard solution, and press the "zero point qualified" button of the electrode after the data is stable.

Slope calibration: Put the electrode into the 4.00 or 9.18/10.00 standard solution, and then press the "slope qualified" button of the electrode after the data is stable.

2. NH4 calibration:

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Date: 2024-07-11
Time: 13:36:52

NH4 Calibration [Back]

NH4 0.00 mg/L Communication -2

Voltage 0.00 mV Temp 0.0 °C

[Start]

The first point 0.000 [First Point Qualified]

The second point 0.000 [Second Point Qualified]

[Home page] [Curve] [History Data] [Calibration] [System]

Firstly, please prepare two suitable standard solutions based on the concentration of the tested water sample. The concentration of the two standard solutions should differ by 10 times. Then clean the electrode with distilled water first, click "Start Calibration", and put it into the first standard solution. Enter the value of the first standard solution, wait for 3 minutes, observe the voltage value, and when the voltage value stabilizes, click "First Qualified". After completion, rinse the electrode with distilled water at least twice and dry it with paper. Put it back into the second standard solution, input the value of the second standard solution, wait for 3 minutes, observe the voltage value, and when the voltage value stabilizes, click "Second qualified" to complete the calibration.

The calibration method for all ion electrodes is the same, please refer to the above method

3.TSS calibration:

BOQU Shanghai Boqu Instrument Co.,Ltd

Date: 2024-07-11
Time: 13:37:08

TSS Calibration [Back]

TSS 0.00 mg/L TSS factor 0.000

Communication -2

[Start]

Input standard buffer value 0.00 [Save]

Set clean period 0.00 [Save] [Manual]

[Home page] [Curve] [History Data] [Calibration] [System]

After entering the calibration interface, place the sensor into a standard solution of known concentration, click "start", enter the value of the standard solution, wait for at least 10 minutes, wait for the value to stabilize, click "save" to complete the calibration.

Scratching settings: Enter the time interval for starting the scraping process, and then click "save". The minimum is 15 minutes, and the maximum is 720 minutes. Click "Manual" and the scraping will immediately start once.

4.COD calibration:

Click “COD Calibration” to enter the COD calibration interface. As shown below:

The screenshot shows the 'COD Calibration' screen. At the top, the company name 'BOQU Shanghai Boqu Instrument Co.,Ltd' is displayed along with the date '2024-07-15' and time '09:33:22'. The main area contains several input fields: 'COD' with a value of '0.00' and unit 'mg/L', 'COD factor' with '0.000', 'Communication' with '-2', and 'Standard Solution Value Input' with '0.00'. There are three buttons: 'Start' (cyan), 'Save' (cyan), and 'Back' (cyan). At the bottom, a navigation bar has five options: 'Home page', 'Curve', 'History Data', 'Calibration', and 'System'.

In the calibration interface, put the COD sensor into standard solution, then click “Start”, waiting for COD value to stable, input standard solution value in “Standard Solution Value Input”, click “Save” to complete the calibration.

In the system settings menu, click "COD Maintenance" to set the COD scraping time. As shown in the following figure:

The screenshot shows the 'COD Setting' screen. At the top, the company name 'BOQU Shanghai Boqu Instrument Co.,Ltd' is displayed along with the date '2024-07-15' and time '09:33:51'. The main area is divided into two sections. The 'Hand Brush' section has three buttons: 'Start the manual wiper', 'Manual scraping once', and 'Reset'. The 'Automatic brush setting' section has a text input field showing '0', and two buttons: 'Save' and 'Reset'. There is also a 'Back' button in the top right. At the bottom, a navigation bar has five options: 'Home page', 'Curve', 'History Data', 'Calibration', and 'System'.

Manual scraping: First click "Start scraping", then click "Manual scraping once", and the electrode scraping will automatically run once. Scratching completed, click "Reset".

Automatic brushing settings: Enter the brushing interval time and click "Save Settings". The shortest time is 15 minutes, and the longest is 1440 minutes per day. If you need to reset, click "Reset" first, and then repeat the above action

Chapter V Common Faults and Handling

No.	Faults	Solutions
1	touch screen no display	Check whether the power supply has electricity; Whether the power supply voltage is normal; Whether the power cord is in good contact;
2	The measurement data varies greatly or there is no data	Check whether the water sample has changed significantly. Whether the sensor connection is normal;

Chapter VI Precautions

When using it for the first time, place the instrument securely, connect the electrodes to the instrument, and then turn on the power. The electrode power supply is 12V, and there are corresponding wiring marks on the instrument. Please pay attention when installing. All electrodes are consumables and need to be cleaned regularly.

Secondly, if the electrode is not in contact with water samples for a long time, the instrument should be powered off. The protective cap of the pH electrode needs to be put back on the electrode, and 3 mol of saturated potassium chloride solution should be added to the protective cap to avoid damage to the electrode.

Chapter VII Multi-parameter MODBUS Communication Protocol

Parameter setting for communication:

- Common baud rate: 19200、9600、4800、2400。Default using 9600
- data bits: 8
- Parity check: no check, odd check, even check. Default is no check
- Stop bit: 1
- Device (instrument) address: The default value of multiple parameters is 5, which cannot be modified by customers

Communication command format:

Data read command:

Address + function code + register start address + number of registers read + CRC check

(hexadecimal)

Example: 05 03 00 00 00 16 C5 80 (read six data)

Site	function code	register start address	number of registers read	CRC check
05	03	0000	00 16	C5 80

data return command:

Site+function Turbiditye+data length+data1+data2.....data11+CRC check (hexadecimal)

Example: 05 03 2C 41 24 CC CD 40 E3 33 33.....+CRC

Site	function code	data length	data1...	Data11	CRC check
05	03	2C	4124CCCD	40 E3 33 33	CRC

Multi parameter data is a floating-point number, sorted as ABCD. And the high and low bytes are not reversed.

For example, the software calculation is as follows:

IEEE-754十六进制数转换十进制浮点数

十六进制数: 40E33333

\$40 \$E3 \$33 \$33

转换

十进制数浮点: 7.09999990463257

The register address is as follows:

Register 40001:Temp

Register 40003:pH

Register 40005:COD

Register 40007:TSS

Register 40009:NH4

Sensor ID

COM2:1:PH , 2:COD , 3:TSS, COM3:1:NH4